Jin Yao

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Education

University of Virginia — Sept. 2023 – Dec. 2027 (Expected)
Ph.D. in Computer Science, GPA: 3.9/4.0, Advisor: Prof. Zezhou Cheng — *Charlottesville, VA, USA*

Shanghai Jiao Tong University — Sept. 2019 – Jun. 2023
B.E. in Artificial Intelligence (Excellent Talent Pilot Class) — *Shanghai, China*

Research Interests

- **3D Computer Vision:** 3D Scene Understanding, Open-Vocabulary 3D Detection, 3D Reconstruction, 3D Generation
- **Physical AI:** Vision-Language-Action Models, World Models, Autonomous Driving, Robotics

Internships

Uber AV Labs, Research Intern — May 2026 – Aug. 2026
Vision-Language-Action Models and World Action Models for Autonomous Driving — *Sunnyvale, CA, USA*

- Built VLGA, the first driving VLA model supervised to reconstruct dense 3D geometry, by adding a geometry expert to a Qwen3-VL Mixture-of-Transformers policy trained with per-pixel pointmap regression against LiDAR [P.1].
- Achieved state-of-the-art open-loop planning among VLA methods on nuScenes, lowering collision rate by 18% relative to prior SOTA, and a state-of-the-art closed-loop driving score on Bench2Drive.
- Developed a video-diffusion-based world action model for autonomous driving and scaled its training to 256 NVIDIA B200 GPUs, proposing a Mixture-of-Transformers design that decouples motion planning from future imagination.
- Introduced future-frame DINO feature world modeling, enriching the model's representations with semantics beyond pixel-level prediction and achieving strong NAVSIM results; manuscript in preparation for CVPR 2027.

Publications

C = Conference, P = Preprint, * = Equal Contribution

[C.1] **J. Yao**, C. Saunders, A. Roy Chowdhury, Z. Cheng. MeshAnyScene: Amodal Mesh Reconstruction from Scene Point Clouds. **3DV'27 conditional accept**.

[C.2] H. Gu*, **J. Yao***, Z. Cheng. Click3R: Interactive Stereo 3D Reconstruction with Sparse Correspondence Clicks. **NeurIPS'26**.

[P.1] **J. Yao**, D. D. Kurra, T. Lampo, Z. Cheng, D. Guo, B. Yaman. VLGA: Vision-Language-Geometry-Action Models for Autonomous Driving. *In Submission.* [Project Page]

[C.3] **J. Yao**, H. Gu, X. Chen, J. Wang, Z. Cheng. Open Vocabulary Monocular 3D Object Detection. **3DV'26**. [Project Page]

[C.4] **J. Yao***, R. M. Redoy*, S. Elbaum, M. B. Dwyer, Z. Cheng. LabelAny3D: Label Any Object 3D in the Wild. **NeurIPS'25**. [Project Page]

[C.5] **J. Yao**, E. Chien, M. Du, X. Niu, T. Wang, Z. Cheng, X. Yue. Machine Unlearning of Pre-trained Large Language Models. **ACL'24 (Main)**. [Code]

[C.6] K. Song, **J. Yao**, R. Jiang, R. Chandra, S. Zhang. Towards Large Language Models that Benefit for All: Benchmarking Group Fairness in Reward Models. **RLC'25**.

[C.7] C. Gong, K. Li, **J. Yao**, T. Wang. TrajDeleter: Enabling Trajectory Forgetting in Offline Reinforcement Learning Agents. **NDSS'25**.

Research Experience

UVA Computer Vision Lab, University of Virginia — Sept. 2023 – Present
Graduate Research Assistant, Advisor: Prof. Zezhou Cheng — *Charlottesville, VA, USA*

- *Interactive 3D Reconstruction:* Proposed Click3R, the first interactive feed-forward stereo reconstruction model, which injects sparse user clicks into DUSt3R via a lightweight adapter; curated a benchmark of hard failure cases, on which Click3R with correspondence clicks delivers large improvements over feed-forward baselines [C.2].
- *Amodal Mesh Reconstruction:* Built MeshAnyScene, which generates complete meshes for partially observed objects in scene scans via Chain-of-Completion decoding and consensus test-time scaling; reduced Chamfer distance by 31% over the best prior method and occluded-surface error by 34% in zero-shot transfer [C.1].
- *Open-Vocabulary 3D Detection:* Formulated open-vocabulary monocular 3D detection and built OVMono3D-LIFT, which lifts Grounding DINO detections to 3D boxes using DINOv2 features and metric depth; achieved state-of-the-art zero-shot AP3D on novel categories with ∼22× less training compute than prior SOTA [C.3].

- *3D Auto-Labeling:* Built LabelAny3D, an analysis-by-synthesis 3D auto-labeling pipeline whose labels achieve 6.4× higher AP3D than prior auto-labeling and nearly double detector AP3D; used it to build COCO3D, an in-the-wild open-vocabulary monocular 3D detection benchmark [C.4].

Teaching

Teaching Assistant, CS 4501 Computer Vision, University of Virginia	Spring 2026
Teaching Assistant, CS 6501 3D Computer Vision, University of Virginia	Fall 2025
Teaching Assistant, CS 4501 Computer Vision, University of Virginia	Spring 2025

Honors and Awards

NeurIPS 2025 Scholar Award	Oct. 2025
Shanghai Jiao Tong University, Outstanding Graduate	Jun. 2023
Shanghai Jiao Tong University, Undergraduate Class C Scholarship	Sept. 2022
Service Outsourcing Innovation and Entrepreneurship Competition, National Second Prize	Aug. 2022

Services

- **Conference Reviewer:** NeurIPS 2026, CoRL 2026, 3DV 2026–2027, AAAI 2026–2027, IJCAI 2026
- **Journal Reviewer:** Transactions on Machine Learning Research (TMLR)

Skills

- **3D Vision:** 3D Object Detection, 3D/4D GS, Image-to-3D Generation, 3D Reconstruction, Mesh Generation
- **Programming & Frameworks:** Python, C++, PyTorch, Hugging Face, PyTorch3D, Open3D, detectron2
- **Tools:** Git, Docker, Linux, LaTeX, Blender, WandB, SLURM, Claude Code, Cursor, Codex
- **Languages:** Mandarin (Native), English (Professional)